

Architectural Reconstruction of the DevSecOps Environment: Analysis of Google Antigravity 2.0 and Gemini 3.5 Flash Integration

The launch of the Google Antigravity 2.0 platform during the Google I/O 2026 developer conference represents a fundamental transition in the design of artificial intelligence-assisted development environments.¹ By moving away from the traditional paradigm focused purely on integrated development interfaces (IDEs) to consolidate an autonomous multi-agent orchestration ecosystem, the architecture redefines the software engineer's interaction with the source code.¹ This report provides a comprehensive and rigorous analysis of the operational capabilities, operating system isolation boundaries, quota dynamics, and storage configurations that characterize this new agentic environment.¹

Execution Engine Metrics: Gemini 3.5 Flash vs. Gemini 3.1 Pro

The execution core of Antigravity 2.0 relies by default on the Gemini 3.5 Flash model, which replaces Gemini 3.1 Pro in this role.⁵ Although both models are built on the multimodal foundation of Google's third generation of native reasoning models, their architectures and performance profiles differ substantially in agentic production environments.⁸

Context Window and Transfer Limits

Both Gemini 3.5 Flash and Gemini 3.1 Pro offer an equivalent input context window of **1,048,576** tokens (1M), complemented by a maximum output generation limit of **65,536** tokens (64K).⁸ However, the operational difference lies in processing efficiency and latency.¹² While Gemini 3.1 Pro is a dense model designed for complex programming logic and high-fidelity mathematical analysis, Gemini 3.5 Flash utilizes a Mixture-of-Experts (MoE) architecture optimized for long-horizon executions and ultra-low latency.⁹

Performance in Agentic and Programming Benchmarks

Laboratory evaluations and independent performance analysis consolidate Gemini 3.5 Flash as an optimal engine for the sequential execution of development tools ¹²:

- **Terminal-Bench 2.1:** Gemini 3.5 Flash reaches an accuracy level of **76.2%**, surpassing previous limits of Gemini 3.1 Pro in the autonomous synthesis and execution of system commands.¹²
- **GDPval-AA:** It records an Elo rating of **1656** points, demonstrating high reliability in the logical self-evaluation of generated code before its commitment to disk.¹²
- **MCP Atlas:** Achieves a success rate of **83.6%**, validating its ability to interact flawlessly with multiple servers under the Model Context Protocol standard.¹²
- **CharXiv Reasoning:** Dominates multimodal logical evaluation with **84.2%**, allowing the agent to interpret network architecture diagrams, UI mockups, and large technical documentation without the need for format conversion.¹²

Regarding output speed, Gemini 3.5 Flash generates an average of **289** tokens per second in standard environments.³ When integrated locally into the Antigravity 2.0 harness, the model leverages hardware acceleration layers that multiply its execution speed by twelve in closed reasoning loops.⁶

This difference in performance and architecture is detailed in the following comparative table ⁸:

Technical Dimension	Gemini 3.5 Flash (Default)	Gemini 3.1 Pro (Legacy)
Network Architecture	Sparse Mixture-of-Experts (MoE) ¹⁴	Dense Reasoning ⁹
Input Context Window	1,048,576 Tokens ⁸	1,048,576 Tokens ¹¹
Output Token Limit	65,536 Tokens ⁸	65,536 Tokens ¹¹
Average Output Speed	≈ 289 tokens/sec (\$12\times\$ local acceleration) ³	$\approx 68 - 73$ tokens/sec ¹⁰
Terminal-Bench 2.1 Accuracy	76.2% ¹²	Surpassed by Gemini 3.5 Flash ¹²

Tool Call Error Rate	Low (Optimized MCP fidelity) ¹²	Moderate ($\approx 1.98\%$ - 5.66%) ¹⁰
Cost per 1M Input Tokens	Reduced (Highly efficient MoE structure) ¹³	$\leq 200K$: \$2.00 / $> 200K$: \$4.00 ¹⁰
Cost per 1M Output Tokens	Reduced (Highly efficient MoE structure) ¹³	$\leq 200K$: \$12.00 / $> 200K$: \$18.00 ¹⁰

Multivariate Analysis of Accelerated Credit Consumption in Gemini 3.5 Flash

Software developers who have integrated version 2.0 of Antigravity into their daily workflows have identified a drastic reduction in the duration of their Google One subscription or corporate AI plan credit balances when using Gemini 3.5 Flash.¹⁴ This phenomenon responds to three technical and economic factors intrinsic to the system's new agentic design²:

Massive Parallelism of Dynamic Sub-agents

The architecture of Antigravity 2.0 promotes the automatic instantiation of specialized sub-agents to tackle complex subtasks concurrently, thus avoiding saturating the main agent's context window.² For example, when faced with a complex request for code refactoring, unit test updates, and API documentation generation, the main agent will split these goals and launch independent sub-agents to execute them in parallel.²

Each sub-agent operates in its own isolated conversational context.² During the proofs of concept conducted at the I/O 2026 conferences, the construction of an experimental operating system core triggered the automatic initialization of up to **93** individual sub-agents and generated more than **2,600** million accumulated tokens in just twelve hours.⁵ Although each individual Flash model interaction is inexpensive, the volume of simultaneous calls drastically accelerates global processing resource consumption.¹³

Cost Duplication Penalty for Large Context

The Vertex AI backend pricing policy imposes a cost penalty for massive conversational sessions.¹⁴ As soon as the context window of an active conversation exceeds the threshold of **250,000** tokens, the model's billing rate for processing input and output tokens automatically doubles.¹⁴

Because Antigravity 2.0 agents continuously index local directories, read version control histories, and collect terminal execution logs, conversational histories cross this **250K** limit with high frequency.¹⁴ Continuing development on a long conversation without segmenting exponentially multiplies credit consumption for each new conversational turn.¹⁴

Fee Consolidation under API Parity

Google has reconfigured the quota system by unifying the old application-specific turn limits into an integrated credit pool based strictly on Vertex AI API prices (**\$3 USD** per million input tokens and **\$9 USD** per million output tokens in the Flex Paygo flexible payment mode).¹⁴ In the previous version of the application, frequent use of Gemini 3.1 Flash did not consume the Pro model quota because the limits were segregated by usage channel.¹⁷

In version 2.0, quota limits for all models have been unified under a conversion rate proportional to their actual cloud processing cost.¹⁸ Thus, for example, if Gemini Flash is eight times cheaper than Gemini Pro on the backend, token consumption in Flash will directly consume the same proportion of the programmer's unified credit pool, eliminating any free and unlimited usage subsidies.¹⁸

Interface Evolutionary Analysis: Legacy Agent Manager vs. Standalone Agent Hub v2.0

The architecture of Antigravity 2.0 completely removes the integrated native code editor interface (the VS Code fork on which version 1.x was based) to offer a standalone desktop application optimized exclusively for monitoring agentic behavior.² The old "Agent Manager" interface of the IDE has been replaced by the new **Agent Hub**, substantially modifying the programmer's operational capabilities¹:

Architectural Feature	Legacy Agent Manager (IDE-Coupled)	Standalone Agent Hub v2.0

Runtime Environment	Rigidly integrated into the Antigravity IDE graphical interface. ²	Standalone desktop application; decoupled from the text editor. ¹
Sub-agent Control	Manual. The programmer had to create, assign roles, and coordinate the workflows of each agent individually. ⁶	Autonomous and Dynamic. The main agent automatically generates dedicated secondary agents and consolidates their results into the main thread. ²
Concurrency Model	Blocking. Compilation or background execution processes froze the agent interaction interface. ²	Decoupled Synchronous/Asynchronous . Tasks are processed in the background independently without interrupting the programmer's conversation. ²
Execution Cycle Control	Natural language text messages to guide and intercept behaviors. ²⁴	Native Execution Control Hooks (JSON Hooks) to modify agentic behavior before and after API calls. ²
Temporal Automation	Passive. Active and immediate user intervention via prompts was required to initiate tasks. ⁶	Scheduled Tasks. Native cron support to automate routine tasks (e.g., nightly dependency analysis). ¹
Context Coordination	Monorepo. Strictly limited to the active working directory of the open window in the IDE. ²	Multi-repo. Ability to coordinate simultaneous operations over multiple directories and local Git repositories. ²

Voice Input Mode	Basic storage. Raw voice recording for subsequent analysis and complete transmission to the cloud model. ²	Direct local transcription based on Gemini Audio models; with automatic removal of pauses and filler words. ²
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Definition of Multi-repo Projects and Isolation Boundaries

In the classic version of Antigravity, an AI agent's logical boundaries were exclusively coupled to a single working folder or development space configured in the code editor.² The launch of version 2.0 introduces the concept of **Projects**, allowing multiple local repositories or folders to be grouped under a single unified agentic context.²

Mechanics of Multi-repo Project Configuration

The initialization and management of multi-repo projects in Antigravity 2.0 is governed by the following technical specifications for file configuration in the user's storage system ²⁸:

1. When creating a new project in the side menu of the Agent Hub interface, the programmer selectively adds multiple local folders or Git repositories using the **Add Folder** option.²⁸
2. The system compiles and consolidates these logical boundaries into the Hub's Protocol Buffers database, located in the binary file
~/.gemini/antigravity/agyhub_summaries_proto.pb.³⁰
3. Project metadata and local paths corresponding to each associated repository are written into individual manifest files stored within the path ~/.gemini/config/projects/.³⁰

Execution Modes and System Permission Limits

The programmer has a hierarchical configuration architecture to establish which files an agent can read and which commands it can execute within a project.²⁷ The system offers three security configurations to delimit access to local files ²⁷:

- **Default Mode (Limited Access):** The agent only has read and write permissions over folders explicitly added to the project in the manifest.²⁷ Any attempt to read outside these limits is immediately denied.²⁷

- **Full Machine Mode:** The agent can inspect and modify files across the host's entire operating system, but the system requires active user confirmation before any write changes are executed outside the workspace.²⁷
- **Unrestricted Mode:** The agent is granted full access to read and write autonomously in any path on the local disk without needing to interact with the user.²⁷

For operations in the operating system console, the programmer can parameterize the following terminal execution directives in the project manifest ²⁷:

- **Interactive Approvals:** Explicit authorization by the programmer is required for each terminal command proposed by the agent before processing.²⁷
- **Request Review Mode:** The agent groups and visually presents planned commands, offering the programmer the option to manually edit commands or force their execution within the safety of the local sandbox.⁴
- **Auto-Accept Mode:** The agent executes all terminal operations automatically and asynchronously, notifying the user only of exceptions and logical blocks.²⁴

Containment via Git Worktrees

To ensure the stability of the programmer's active directory, Antigravity 2.0 incorporates native support for **Git Worktrees**.²⁶ When initializing a conversational agent with the **New Worktree** directive, the Antigravity runtime automatically creates an isolated Git working copy in a background temporary directory.²⁶

The agent executes all its tasks (compilations, refactorings, dependency changes, and unit tests) within this isolated worktree.²⁶ This allows the programmer to continue editing their work branch simultaneously in their traditional editor, eliminating cross-contamination and accidental merge conflicts while agents operate concurrently.²

Linux Systems Deployment Guide: Clean Installation and Reversion Procedures

Updating to Antigravity 2.0 on Linux-based distributions (such as Linux Mint, Ubuntu, or Debian) is often affected by APT repository conflicts.³⁴ Since Google has decoupled the IDE and the Agent Manager into two independent binaries, the operating system package repositories may experience architecture mismatches.³⁴

This causes ordinary updates (apt upgrade) to download and install obsolete and conflicting versions of the application (such as legacy version 1.23.2) over new system folders.³⁴ This mismatch triggers a startup redirect loop (Redirect Loop), caused by legacy process managers trying to take control of the new decoupled interfaces.³⁴

Purge and Clean Installation Procedure

To resolve this state of inconsistency and consolidate a clean side-by-side installation of both tools, the following commands must be executed in the Linux terminal ³⁴:

1. Stop Frozen Runtimes and Debug Conflicting Files

All traces of obsolete installations must be removed to prepare the operating system ³⁴:

Bash

Shell

```
# Force close any orphaned Antigravity threads or processes
pkill -f antigravity

# Completely remove the corrupt package from the system's APT
repositories
sudo apt purge antigravity -y

# Remove obsolete shortcuts and update the window manager database
sudo rm -f /usr/share/applications/antigravity.desktop
sudo update-desktop-database

# Clear cache folders and legacy environment configurations
rm -rf ~/.antigravity
```

2. Acquire and Deploy the New Binaries Portably

Portable binary packages are downloaded directly from official Google servers to avoid synchronization failures of ordinary repositories ³⁴:

Bash

Shell

```
# Create a secure local directory to house portable applications
mkdir -p ~/Apps/AntigravitySuite
cd ~/Apps/AntigravitySuite
```



```
# Download and extract the standalone Agent Manager 2.0 package for
Linux x64
wget -O Antigravity-x64.tar.gz
"https://antigravity.google/download/linux/Antigravity-x64.tar.gz"
tar -xzf Antigravity-x64.tar.gz
rm Antigravity-x64.tar.gz

# Download and extract the decoupled version of the Antigravity IDE
for Linux x64
wget -O AntigravityIDE-x64.tar.gz
"https://antigravity.google/download/linux/AntigravityIDE-x64.tar.g
z"
tar -xzf AntigravityIDE-x64.tar.gz
rm AntigravityIDE-x64.tar.gz
```

3. Grant Execution Attributes to Portable Binaries

To avoid security blocks from the operating system kernel, execution permissions are assigned to the extracted binaries ³⁴:

Bash

```
Shell
chmod +x ~/Apps/AntigravitySuite/Antigravity-x64/antigravity
chmod +x ~/Apps/AntigravitySuite/Antigravity-IDE/antigravity-ide
```

4. Manually Register the New Launchers in the System

New desktop shortcuts must be configured within the user's local path to allow parallel and independent execution of both tools ³⁴:

Bash

```
Shell
# Create the desktop launcher for Antigravity Agent Manager 2.0
cat << 'EOF' >
~/local/share/applications/antigravity-manager.desktop
Version=2.0.0
```

```

Type=Application
Name=Antigravity Agent Manager 2.0
Comment=Autonomous Multi-agent Orchestration Console
Exec="/home/USERNAME/Apps/AntigravitySuite/Antigravity-x64/antigravity" %F
Icon=/home/USERNAME/Apps/AntigravitySuite/Antigravity-x64/resources/app/icon.png
Terminal=false
Categories=Development;IDE;
StartupWMClass=antigravity
EOF

# Create the desktop launcher for the decoupled Antigravity IDE
cat << 'EOF' > ~/.local/share/applications/antigravity-ide.desktop
Version=2.0.0
Type=Application
Name=Antigravity IDE
Comment=AI Agent-Assisted Development Environment
Exec="/home/USERNAME/Apps/AntigravitySuite/Antigravity-IDE/antigravity-ide" %F
Icon=/home/USERNAME/Apps/AntigravitySuite/Antigravity-IDE/resources/app/icon.png
Terminal=false
Categories=Development;IDE;
StartupWMClass=antigravity-ide
EOF

# Adapt USERNAME environment variables to the current operating system user
sed -i "s|/home/USERNAME|$HOME|g"
~/.local/share/applications/antigravity-manager.desktop
sed -i "s|/home/USERNAME|$HOME|g"
~/.local/share/applications/antigravity-ide.desktop

# Validate permissions and force reload of host desktop application engine
chmod +x ~/.local/share/applications/antigravity-manager.desktop
chmod +x ~/.local/share/applications/antigravity-ide.desktop
update-desktop-database ~/.local/share/applications

```

Emergency Rollback Protocol and State Recovery

If the new portable environment presents logical incompatibilities, connection blocks in the

Linux Subsystem for Windows (WSL), or graphical rendering failures of the standalone application, clean version 1.23.2 can be restored ³⁶:

1. Debug Environment 2.0 and Reinstall Stable Legacy Version

New Antigravity 2.0 directories are removed and the verified legacy version is downloaded ³⁴:

Bash

Shell

```
# Force close processes and delete version 2.0 launchers
pkill -f "antigravity"
rm -f ~/.local/share/applications/antigravity-manager.desktop
rm -f ~/.local/share/applications/antigravity-ide.desktop

# Create a storage path and install the original 1.23.2 Linux x64 package
mkdir -p ~/Apps/AntigravityLegacy && cd ~/Apps/AntigravityLegacy
wget
"https://edgedl.me.gvt1.com/edgedl/release2/j0qc3/antigravity/stabl
e/1.23.2-4781536860569600/linux-x64/Antigravity.tar.gz"
tar -xzf Antigravity.tar.gz
rm Antigravity.tar.gz
chmod +x ~/Apps/AntigravityLegacy/antigravity
```

2. Permanently Block Automatic Update Mechanism

To prevent the classic version from updating to version 2.0 again on its first background login, a block directive must be added to the user's configuration file ³⁷:

Bash

Shell

```
# Ensure the existence of the Antigravity user configuration
directory
mkdir -p ~/.config/Antigravity/User/

# Inject "update.mode": "none" into settings.json
if [ -f ~/.config/Antigravity/User/settings.json ]; then
    python3 -c '
```

```

import json, os;
path =
os.path.expanduser("~/config/Antigravity/User/settings.json");
data = json.load(open(path)) if os.path.exists(path) else {};
data["update.mode"] = "none";
json.dump(data, open(path, "w"), indent=4)
'
else
    echo '{"update.mode": "none"}' >
~/config/Antigravity/User/settings.json
fi

```

3. Synchronize Conversation History and Development Environment Extensions

To recover the programmer's work state without losing installed extensions or personalized development space configurations, profile files must be migrated ³⁹:

Bash

```

Shell
# Generate the structural path for loading legacy add-ons in the
IDE
mkdir -p ~/.antigravity-ide/extensions/

# Clone user custom extensions to the classic IDE folder
if [ -d ~/.antigravity/extensions/ ]; then
    cp -r ~/.antigravity/extensions/*
~/.antigravity-ide/extensions/
fi

# Force the IDE to index clean state by deleting previous extension
cache
rm -f ~/.antigravity-ide/extensions/extensions.json

```

Portable Distribution Model and Security Isolation Mechanisms

Google's decision to release Antigravity 2.0 as a portable and self-contained desktop application package, instead of using traditional system installers, responds to a strategy of

vulnerability mitigation and logical software stability.² This model avoids overwriting system libraries, prevents logical conflicts during concurrent task execution, and provides a uniform abstraction layer to isolate the AI agent from the host operating system.⁴

Isolation Layers and Active Sandbox Engines

When an agent within the Agent Hub or Antigravity CLI interface executes local commands in the development command console, the host operating system does not process these calls directly.⁴ Instead, the Antigravity system encapsulates and isolates them using native security mechanisms from each operating system's kernel ⁴:

- **In Linux (nsjail):** Antigravity relies on the nsjail sandboxing utility.⁴ This engine isolates agent command processes within independent namespaces (PID, UTS, mount, and network).⁴ nsjail restricts agent write access strictly to configured project folders, mounting the rest of the OS filesystem in read-only mode.⁴ This prevents accidental actions or logical attacks from destroying the OS or accessing user data.⁴
- **In macOS (Seatbelt / sandbox-exec):** The system integrates with Apple's internal isolation infrastructure, known as Seatbelt.⁴ The runtime generates dynamic real-time sandbox profiles that prevent agent-initiated subprocesses from accessing unauthorized network sockets or modifying system files outside the development workspace.⁴
- **In Windows (AppContainer):** The system executes commands within low-privilege security containers using Windows AppContainer isolation profiles.³² This prevents the development agent from interacting with the host's main administrative processes.³²

In addition to this command console isolation, browser integration is protected under the Chromium sandbox architecture.²⁷ When executing the web navigation agent using the `/browser` command, the browser opens a debugging session via the Chrome DevTools protocol, requiring active debugging permissions before any automated session can begin.²⁷ At the local Linux file system level, the gravity-cli tool manages editor and Google Chrome paths using the `dpkg-divert` command, ensuring that `--no-sandbox` security directives required in specific configurations remains controlled against possible host OS updates.⁴³

Vulnerability Analysis: Prompt Injection and Sandbox Escape in `find_by_name`

The critical importance of this isolation architecture was evidenced by a security flaw reported by Pillar Security researchers in early 2026, which exposed how a command injection attack in native AI development tools could compromise local systems if strict sandboxing was not applied.⁴¹

The Exploitation Mechanism

The attack vector affected Antigravity's native `find_by_name` tool.⁴¹ This function helps the agent locate files on the programmer's disk by internally encapsulating the `fd` terminal search tool.⁴¹ Because search was classified as a native tool call and not a direct command executed in the shell terminal, the system executed this code automatically before validating OS Secure Mode security restrictions.⁴⁴

The exploitation benefited from two flaws in Antigravity's internal code⁴⁴:

1. **Lack of Input Validation:** The tool's `Pattern` parameter accepted text strings without validating or filtering control characters or commands.⁴⁴
2. **Omission of Argument Terminator:** The application constructed the call to the `fd` binary without including the standard `--` argument terminator before passing the user's search string.⁴⁴

An attacker could inject malicious text into an ordinary file in a public code repository (e.g., a `README.md` file).⁴¹ Upon ordering the agent to analyze the repository, the prompt injection forced the model to perform a local search with the `Pattern` parameter modified to include the `-X` (exec-batch) command of the `fd` tool.⁴⁴ In this way, the system call was executed as follows:

```
Operación en el Host = fd [opciones] -X /tmp/malicious_payload.sh [ruta_búsqueda]
```

Because the development agent had permissions to modify local files, the attack wrote a malicious script to disk and then forced its execution outside the traditional sandbox environment through the native search command.⁴¹

Containment and Architectural Resolution

Google fixed this command injection flaw in late February 2026 by applying a structured two-level solution⁴⁴:

- **Forced Parameter Termination:** The agent harness was modified to automatically prepend the `--` terminator before any user-supplied string, forcing the `fd` utility to treat the input strictly as a literal search string and not as system execution flags.⁴⁴
- **Unified Execution Sandbox:** The application architecture was modified so that all internal native tools (including searches, file readers, and AST utilities) run strictly within the same limited context of the kernel-level sandbox (`nsjail/Seatbelt`), neutralizing any possibility of process escape to the host system.⁴

Token Consumption Optimization via Context Caching

and Local Intelligence

One of the most effective technologies for reducing operating costs in Antigravity 2.0 is the implementation of **Context Caching** in the Google Cloud, complemented by local semantic analysis from the Agent Hub.¹⁴

Cloud Context Cache Financial Structure

Under the traditional linguistic model processing, the agent must read and re-analyze the entire conversation history, project structure, and indexed files in each new conversational turn.¹⁴ This causes an exponential growth in processed tokens per second, drastically increasing the cost of routine programming tasks.¹⁴

Cloud Context Caching technology allows the working context window (such as the code repository index, database schemas, and project documentation) to be stored directly in the active memory of Google's cloud processing server.¹⁴ This significantly alters the development model billing rates¹⁰:

- **Cache Write:** Initial storage of the user's codebase has a cost of **\$0,375** per million tokens processed.¹⁰
- **Cache Read:** Each question turn or prompt sent to the agent reads data directly from the cloud cache at a reduced rate of **\$0,20** per million tokens (or **\$0,40** if cached context exceeds **200.000** tokens).¹⁰

This optimization reduces billed input tokens by approximately **80% – 90%** compared to the ordinary non-cached input rate (**\$2,00** per million tokens under a processing volume lower than **200K**).¹⁰

Coordination via Local Intelligence Engines

To prevent small file modifications from completely invalidating the cloud context cache (which would force a rewrite of the entire codebase and incur additional costs), Antigravity 2.0 uses a local project indexing and hashing system⁴⁵:

1. **Local AST Indexing:** During workspace initialization processes (agyhub_refresh), the Antigravity client uses local code tree parsers (based on GitNexus and Tree-sitter parsing libraries) to map system calls, logical flows, and dependency call graphs.⁴⁵
2. **Hash Calculation and Semantic Diffs:** The software locally calculates the logical footprint (hash) of each functional code block.⁴⁵

3. **Incremental Cache Update:** When editing code, the local Agent Hub identifies exactly which semantic block has changed.⁴⁵ In the next turn, the system sends an incremental update request to the cloud cache, replacing only the modified block instead of invalidating and re-indexing the entire file repository.¹⁴

This architecture also allows basic semantic planning tasks to be derived to smaller, more efficient local models (such as running Gemma via Ollama), resorting to Gemini 3.5 Flash in the cloud only for complex long-horizon operations, thus optimizing the project cost balance.⁴⁶

The Critical Impact of Classic (Legacy) Runtime Permanence

A programmer's decision not to update their tools and continue working on classic versions of the Antigravity application (prior to version 2.0) carries immediate risks of blocking and inactivity in their daily work environment.¹⁴ This penalty does not respond to programmed incompatibility, but rather to the interaction of old development environments with the new usage limit system of Google servers.²⁰

The Dual Quota Mechanism: Sprint Limit and Marathon Baseline

To avoid saturating processing servers against continuous agentic code refactoring executions, Google implemented a logical limit system structured in two simultaneous layers²⁰:

- **Sprint Limit (Immediate Capacity):** It is a short-term usage window that grants the programmer a balance of **250** processing units.²⁰ Each unit consumed by agent operations is automatically recovered exactly five hours after the exact moment of its use, offering a short-duration rolling quota flow optimal for short sessions of assisted programming.²⁰
- **Marathon Baseline (Hard Weekly Quota):** It is a superior hierarchical control directive, designed as a rigid weekly limit consolidated at **2,800** processing units.²⁰ Unlike the five-hour quota, this metric operates on a seven-day calculation window.²⁰

If the accumulated sum of the programmer's operations exhausts the 2,800 units in an interval of less than seven days, the platform backend automatically activates a rigid security restriction that completely disables calls to the model.²⁰ This blocking state invalidates and suspends the resetting of the five-hour rolling quota.²⁰

Because the programmer's panel in the classic environment usually exclusively shows the five-hour quota timer, a recurring state of confusion is generated: the programmer expects the service to be restored at the end of the timer countdown, but the system maintains the block due to the exhaustion of the Marathon weekly quota balance.²⁰

Quota Consumption by Multimodal Element Processing

Ordinary agent-assisted development tasks consume processing logical resources at an accelerated rate when the agent interacts with graphic elements or multimedia control ²⁰:

- **Images and Screenshots:** Processing a web browser screenshot to perform frontend component debugging tasks consumes a base quota rate of **258** tokens if the image dimensions are less than **512 x 512**. ²⁰ If dimensions are larger, the image is divided into pixel grids, billing independently **258** tokens for each grid resulting from fragmentation. ²⁰
- **Video Capture Analysis:** Analyzing videos of graphic components or defective user flows of the user's application consumes approximately **263** tokens for each second of video recording processed by the model. ²⁰
- **Voice Instructions:** Sending instructions via audio messages consumes a quota of **32** tokens for each second of sound processed. ²⁰

Causes of Quota Saturation in Classic Customers

The combination of these usage metrics with the technical limitations of classic application versions causes the exhaustion of the programmer's weekly quota in a few hours due to the following design flaws ²:

1. Full Codebase Transmission Without Segmentation

Lacking support for task division into independent sub-agents and a local semantic diffs engine, the classic client transmits the entire local code repository and complete chat history in each conversation turn. ² This immediately saturates the conversational context window and exhausts the 2,800 weekly units of the Marathon quota in a single interactive work session. ¹⁴

2. Audio Transmission Without Compression or Local Processing

In the previous version of Antigravity, the system collected the programmer's voice in raw audio files and sent them directly to the Vertex AI cloud, consuming massive multimedia input tokens (at a rate of 32 tokens per second of audio). ² In version 2.0, the Agent Hub processes and transcribes the voice locally using specialized models on the host, sending only the clean text prompt of filler words and pauses to the servers, thus reducing the cost of multimedia tokens to zero. ²

3. Quota Authentication Errors due to Backend Migration

The billing and licensing backend of Google has completed the transition from the old local application-based quota allocation model to the new unified credit system of Google One and Vertex AI accounts.⁴⁸

Antigravity classic clients and versions are not compatible with the new backend system's license verification protocol.⁴⁸ When attempting to process command flows, the lack of a correct license signature forces the servers to reject the call or bill the transaction applying generic unoptimized rates, immediately blocking the developer's account with false exhausted quota warnings.¹⁴

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